

Lec 2 Linear Model Selection

Objectives

At the end of this lecture you should be able to:

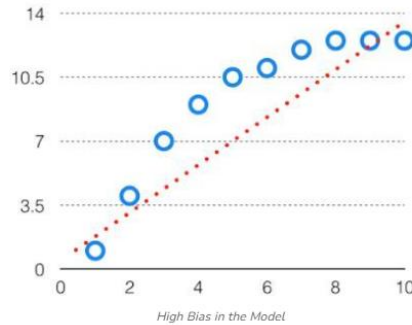
- Understand Bias-Variance Trade Off, overfitting and underfitting.
- Apply the methods of variable selection to see which variables should be kept in the model.
- Learn to build regression models using forward, backward, and stepwise regression.

Bias-Variance Trade Off

- It is important to understand prediction errors (*bias* and *variance*) when it comes to accuracy in any model.
- There is a tradeoff between a model's ability to minimize bias and variance which is referred to as the best solution for selecting a value of *Regularization* constant.
- A proper understanding of these errors would help to avoid the *overfitting* and *underfitting* of a data set while training the algorithm.

Bias

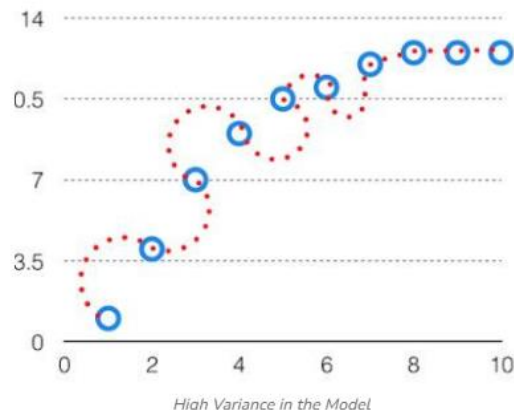
- The bias is known as the difference between the prediction of the values by the model and the correct value.
- Being high in biasing gives a large error in training as well as testing data.
- It is recommended that an algorithm should always be low-biased to avoid the problem of underfitting.
- By high bias, the data predicted is in a straight line format, thus not fitting accurately in the data in the data set. Such fitting is known as the Underfitting of Data.
- This happens when the hypothesis is too simple or linear in nature.
- Refer to the graph given below for an example of such a situation.



In such a problem, a hypothesis looks like follows. $h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$

Variance

- The variability of model prediction for a given data point which tells us the spread of our data is called the variance of the model.
- The model with high variance has a very complex fit to the training data and thus is not able to fit accurately on the data which it hasn't seen before.
- As a result, such models perform very well on training data but have high error rates on test data.
- When a model is high on variance, it is then said to as **Overfitting of Data**.
- Overfitting is fitting the training set accurately via complex curve and high order hypothesis but is not the solution as the error with unseen data is high.
- While training a data model variance should be kept low.
- The high variance data looks as follows.



In such a problem, a hypothesis looks like follows.

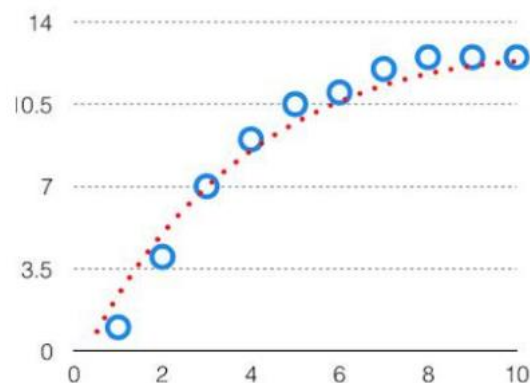
$$h_{\theta}(x) = g(\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4)$$

Bias Variance Tradeoff

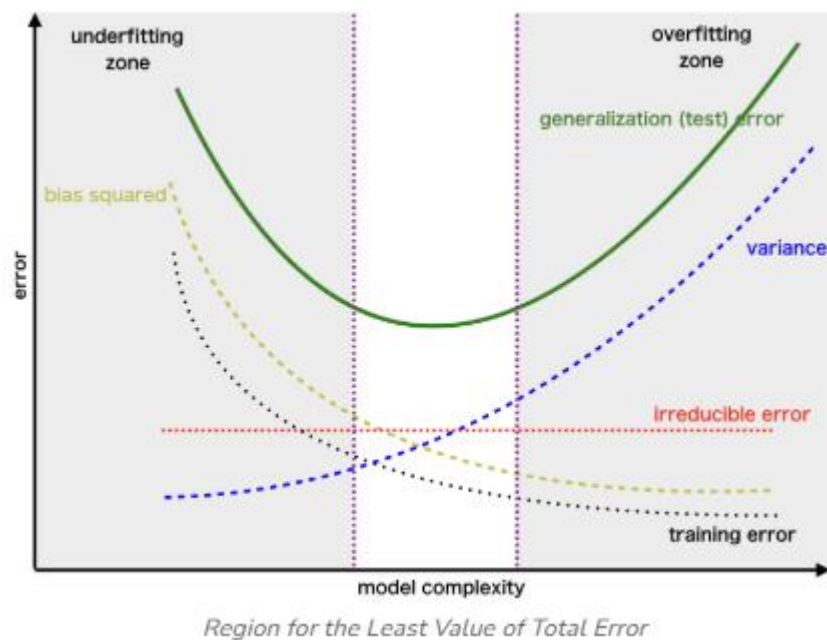
- Bias Variance Trade-off refers to the balance between bias and variance, which affect predictive model performance.
- Finding the right tradeoff is crucial for creating models that generalize well to new data.

$$MSE = \text{Bias}^2 + \text{Variance}$$

- For the graph, the perfect tradeoff will be like this.



The error to complexity graph to show trade-off is given as



- Model Complexity (simpler models on the left, more complex models on the right).

- Error (can be Training Error, Test Error, Bias, or Variance).

Left Side (Underfitting - High Bias, Low Variance)

- Low model complexity.
- High bias: Model makes strong assumptions and oversimplifies patterns.
- Low variance: Model predictions are stable across different datasets.
- Both training and test errors are high (poor performance on all data).

Middle (Optimal Trade-off - Balanced Bias and Variance)

- The ideal point where test error is minimized.
- Balanced bias and variance (good generalization).
- Model complexity is sufficient to capture patterns without overfitting.

Right Side (Overfitting - Low Bias, High Variance)

- High model complexity (high-degree polynomial regression).
- Low bias: Model captures training data patterns very well.
- High variance: Model is sensitive to small changes in the training data and performs poorly on new data.
- Training error is very low, but test error increases (poor generalization).

Why might we want to use another fitting procedure instead of least squares?

Alternative methods often give:

- Better prediction accuracy
- Better model interpretability

Prediction Accuracy

When Does Least Squares Work and When Does It Fail?

The true relationship between Y and the predictors is approximately linear. In this case, least squares estimates have *low bias*.

Best-case for OLS:

$n \gg p$ (number of observations much larger than number of predictors)

- Least squares also has low variance

- Performs very well on test (new/unseen) observations

Problem 1: When n is not much larger than p

- High variability in the least squares fit
- Overfitting occurs
- Poor predictions on future/ test observations

Problem 2: When $p > n$ (more predictors than observations)

- No unique least squares solution exists
- Infinitely many possible coefficient sets
- All of them give zero training error (perfect fit on training data)
- But extremely high variance
- very poor test set performance

Model Interpretability

Why We Need Simpler, Sparser Models

- In multiple regression, many predictors are often irrelevant (not truly associated with the response variable Y)
- Including irrelevant variables causes:
 - Unnecessary complexity in the model
 - Harder to interpret which variables really matter
- Solution: Remove irrelevant predictors completely
 - Set their corresponding coefficients $\beta = 0$ exactly
 - Results in a simpler, more interpretable model (sparser model)
- Problem with OLS:
 - Extremely unlikely to produce any coefficient exactly zero
 - Almost all predictors stay in the model, even if their effect is zero

Alternatives to Least Squares

There are many classical and modern alternatives to fitting the linear model using OLS. We focus on **three important classes** of methods:

1. Subset Selection

- Identify a subset of the p predictors believed to be related to the response.
- Fit a least squares model on this reduced set of variables.
- Goal: Simpler model with only relevant predictors.

2. Shrinkage (Regularization)

- Fit a model using all p predictors.
- Shrink (constrain) the estimated coefficients toward zero compared to least squares.
- Reduces variance (improves prediction accuracy).

- Depending on the method, some coefficients may become exactly zero
- performs variable selection too.

3. Dimension Reduction

- Project the p predictors into a lower-dimensional subspace (M dimensions, where $M < p$).
- Fit the model using these M reduced components instead of the original predictors.
- Reduces complexity and handles multicollinearity/high dimensionality.

Multi-Collinearity

- Multicollinearity occurs when predictor variables are highly correlated, leading to unstable OLS estimates.
- A severe case is when $p > n$ (more predictors than observations), making standard OLS regression infeasible. (**High-dimensionality**)
- OLS estimators are unbiased, but this is not helpful under multicollinearity since they have large variances and are unstable.

Example: The following Hald cement data shown on the heat evolved in calories (y) during hardening of cement per gram for 13 samples of cement and the percentages of their four ingredients: tricalcium aluminate (x_1), tricalcium silicate (x_2), tetracalcium aluminoferrite (x_3), and dicalcium silicate (x_4). Obviously, the four percentages add up to 100%, except for rounding errors and any impurities. So these data are structurally multicollinear.

No.	x_1	x_2	x_3	x_4	y	No.	x_1	x_2	x_3	x_4	y
1	7	26	6	60	78.5	8	1	31	22	44	72.5
2	1	29	15	52	74.3	9	2	54	18	22	93.1
3	11	56	8	20	104.3	10	21	47	4	26	115.9
4	11	31	8	47	87.6	11	1	40	23	34	83.8
5	7	52	6	33	95.9	12	11	66	9	12	113.3
6	11	55	9	22	109.2	13	10	68	8	12	109.4
7	3	71	17	6	102.7						

- The correlation matrix for these data is shown below.

$$\begin{matrix} & x_1 & x_2 & x_3 & x_4 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} & \begin{bmatrix} 1 & 0.229 & -0.824 & -0.245 \\ 0.229 & 1 & -0.139 & -0.973 \\ -0.824 & -0.139 & 1 & 0.030 \\ -0.245 & -0.973 & 0.030 & 1 \end{bmatrix} \end{matrix}$$

We notice that two correlations, $\text{Corr}(x_1, x_3) = -0.824$ and $\text{Corr}(x_2, x_4) = -0.973$, are large and negative.

- The regression output for the cement data is shown in the R output below.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	62.4054	70.0710	0.891	0.3991
x1	1.5511	0.7448	2.083	0.0708 .
x2	0.5102	0.7238	0.705	0.5009
x3	0.1019	0.7547	0.135	0.8959
x4	-0.1441	0.7091	-0.203	0.8441

Residual standard error: 2.446 on 8 degrees of freedom
 Multiple R-squared: 0.9824, Adjusted R-squared: 0.9736
 F-statistic: 111.5 on 4 and 8 DF, p-value: 4.756e-07

We see that all four regression coefficients are nonsignificant, whereas the overall F-statistic is highly significant

Subset Selection

In this section we consider some methods for selecting subsets of predictors. These include:

1. best subset selection
2. stepwise model selection

1. Best Subset Selection

- Best subset selection is a method used in regression modeling to identify the best set of predictors that provide the most accurate predictions.
- Given p predictor variables $(x_1, x_2, \dots, x_p)$, this method evaluates all possible subsets of these variables and selects the best one based on a chosen criterion.
- Each predictor can either be included or excluded, so with p predictors, there are 2^p models .
- For example, if $p = 3$, the possible models are:

- + Model 1: the null model, which has no predictors
- + Model 2: x_1
- + Model 3: x_2
- + Model 4: x_3
- + Model 5: x_1, x_2
- + Model 6: x_1, x_3
- + Model 7: x_2, x_3
- + Model 7: x_1, x_2, x_3
- + Total: $2^3 = 8$

Model Selection Criteria

- Sometimes we have a set of possible models and we want to choose the best model. A p -variable model consists of a subset of size p from $x_1, \dots, x_m$.
- Model selection methods help us choose a good model.
- The most common model selections methods are:
 - R_p^2 Criterion
 - Adjusted R_p^2 Criterion
 - Mallows' C_p Criterion
 - AIC_p and BIC_p

R_p^2 Criterion: This criterion compares models based on their R^2 values, which are measures of the goodness of fit.

Adjusted R_p^2 Criterion: It is defined as

$$R_{adj,p}^2 = 1 - \frac{SSE_p/[n-(p+1)]}{SST/(n-1)}.$$

$R_{sdj,p}^2$ reaches a maximum when MSE_p reaches a minimum at some p .

Mallows' C_p Criterion: Mallows' (1973) C_p statistic is an approximately unbiased estimate of the sum of the standardized mean squared error of prediction (MSEP) of all n observations (except for a constant). It is given by

$$C_p = \frac{SSE_p}{\hat{\sigma}^2} + 2(p + 1) - n.$$

where $\hat{\sigma}^2$ is an unbiased estimate of σ^2 . The goal is to minimize C_p . Usually, the MSE for the full model is used for $\hat{\sigma}^2$, i.e. $\hat{\sigma}^2 = \text{MSE}_m = \text{SSE}_m/[n - (m + 1)]$. This gives

$$C_m = \frac{\text{SSE}_m}{\text{MSE}_m} + 2(m + 1) - n = n - (m + 1) + 2(m + 1) - n = m + 1.$$

AIC_p and BIC_p criteria:

AIC_p stands for Akaike's information criterion and is given by

$$\text{AIC}_p = n \ln \text{SSE}_p + 2(p + 1) - n \ln n.$$

Minimizing AIC_p is equivalent to maximizing the expected information in a model subject to a penalty term for the number of variables in the model.

BIC_p stands for (Schwarz's) Bayesian information criterion and is given by

$$\text{BIC}_p = n \ln \text{SSE}_p + (p + 1) \ln n - n \ln n.$$

Example: (Hald Cement Data: Best Subsets Regression) The values of the above subset selection criteria for the Hald cement data are shown in the following table. For each p , the two best models (in terms of the largest R_p^2) are shown.

Values of the criteria for two best subsets of each size for the Hald cement data

p	Variables	$R_p^2(\%)$	$R_{\text{adj},p}^2(\%)$	C_p	AIC _p	BIC _p
1	x_4	67.45	64.50	138.75	58.85	59.98
	x_2	66.62	63.60	142.49	59.18	60.31
	x_1, x_2	97.87	97.44	2.678	25.42	27.11
	x_1, x_4	97.25	96.69	5.503	28.75	30.44
	x_1, x_2, x_4	98.23	97.64	3.018	24.97	27.23
	x_1, x_2, x_3	98.23	97.64	3.042	25.01	27.27
4	x_1, x_2, x_3, x_4	98.24	97.36	5.0	26.94	29.76

We see that the $R_{\text{adj},p}^2$, C_p , and the BIC_p criteria choose $\{x_1, x_2\}$ as the best model, whereas the AIC_p criterion chooses $\{x_1, x_2, x_4\}$ as the best model, which was also

chosen by lasso regression. However, its AIC_p is only marginally smaller than that of the $\{x_1, x_2\}$ model.

We illustrate the calculation of the criteria for the $\{x_1, x_2\}$ model. For this model, $SST = 2715.76$, $SSE(x_1, x_2) = 57.90$, and $MSE(x_1, x_2) = 5.79$. For the full model $SSE(x_1, x_2, x_3, x_4) = 47.86$, and $MSE(x_1, x_2, x_3, x_4) = 5.98$. Therefore,

$$R_p^2 = 1 - \frac{57.90}{2715.76} = 97.87\%, \quad R_{adj,p}^2 = 1 - \frac{5.79}{2715.76/12} = 97.44\%,$$

$$C_p = \frac{57.90}{5.98} + 2(2 + 1) - 13 = 2.678,$$

$$AIC_p = 13 \ln \left(\frac{57.90}{13} \right) + 2(2 + 1) = 25.418,$$

and $BIC_p = 13 \ln \left(\frac{57.90}{13} \right) + (2 + 1) \ln 13 = 27.113.$

2. Stepwise Regression

- One of the frequently asked questions in multiple regression model development is how to select variables or features to build the model, especially when the data set has a large number of independent variables.
- In this section we will discuss different approaches that can be used for developing multiple regression models by selecting variables automatically.
- Such procedures ensure that only statistically significant variables at a significance value of α are included in the model.
- There are several criteria used for variable selection to develop MLR models; in this section we discuss variable selection based on partial F-test.

Forward Selection

Forward selection starts from the null model and enters variables one at a time. The following steps are used in building regression model using forward selection method.

1. Start with no variables in the model. Calculate the correlation between dependent and all independent variables.
2. Develop simple linear regression model by adding the variable X_i for which the correlation coefficient is highest with the dependent variable.
3. Add the variable X_j from step 2 with smallest p-value based on partial F-test.

4. Repeat step 3 till the smallest p-value based on partial F-test is greater than α .

Backward Elimination Procedure

backward starts from the full model and removes variables one at a time.

1. We start with a multiple regression model with all variables.
2. Remove one variable at a time repeatedly from the model in step 1 and create a reduced model (say model 2). Perform a partial F-test between the models in step 1 and step 2.
3. Remove the variable with largest p-value (based on partial F-test) if the p-value is greater than the significance α (or the F-value is less than the critical F-value).
4. Repeat the procedure till the p-value becomes less than α or there are no variables in the model for which the p-value is greater than α based on partial F-test.

Stepwise Regression

- Stepwise regression is a combination of forward selection and backward elimination procedure.
- In this case, we set the entering criteria (α) or f_{in} for a new variable to enter the model based on the smallest p-value of the partial F-test and removal criteria (β) or f_{out} for a variable to be removed from the model if the p-value exceeds a pre-defined value based on the partial F-test ($\alpha < \beta$).
- For example, we may use $\alpha = 0.05$. If the smallest p-value based on the partial F-test is less than α then the variable will be entered and if the p-value is greater than $\beta = 0.10$, then we will remove the variable from the equation.
- At each step a variable is either entered into the model or removed from the model.
- The first variable to be added to the model is the one that has the highest correlation with the dependent variable provided the p-value corresponding to that variable is less than the significance value α .

Example: (Hald Cement Data: Stepwise Regression) In this example, we will illustrate the stepwise regression algorithm given above for the Hald cement data. We will set $f_{in} = f_{out} = 4.0$.

Step 1: At Step 1, we decide which variable to enter in the model first. Since there are no variables in the model to begin with, the maximum partial F -statistic corresponds to the maximum absolute bivariate correlation coefficient $|r_{yx_j}|$. The four correlation coefficients are

$$r_{yx_1} = 0.7307, r_{yx_2} = 0.8163, r_{yx_3} = -0.5347, r_{yx_4} = -0.8213,$$

and the corresponding error sum of squares (SSEs) are

$$\begin{aligned} \text{SSE}(x_1) &= 1265.7, \text{SSE}(x_2) = 906.1, \\ \text{SSE}(x_3) &= 1939.3, \text{SSE}(x_4) = 883.9. \end{aligned}$$

Also, $\text{SST} = \text{SSE}(\emptyset) = 2715.76$ from the analysis of variance table, where $\emptyset$ denotes a null model with only the intercept term. The maximum absolute bivariate correlation coefficient is $|r_{yx_4}|$, which corresponds to the minimum SSE (since it reduces the SSE most from the null model) or $\max F_{\text{in}}$. Using the extra SS method, we calculate the F_{in} for x_4 :

$$F_{\text{in}} = \frac{\text{SSE}(\emptyset) - \text{SSE}(x_4)}{\text{MSE}(x_4)} = \frac{2715.76 - 883.9}{883.9/(13-2)} = 22.745 > 4.0.$$

So x_4 enters the model.

Step 2: At Step 2, we choose between x_1, x_2 , and x_3 to decide which variable, if any, enters the model next. The variable that gives the smallest SSE will be the candidate. The three SSE values are as follows:

$$\text{SSE}(x_1, x_4) = 74.8, \text{SSE}(x_2, x_4) = 868.9, \text{SSE}(x_3, x_4) = 175.7.$$

Since adding x_1 gives the smallest SSE, it is the candidate for the entry into the model. To test if it meets the entry criterion, we calculate its F_{in} statistic:

$$F_{\text{in}} = \frac{\text{SSE}(x_4) - \text{SSE}(x_1, x_4)}{\text{MSE}(x_1, x_4)} = \frac{883.9 - 74.8}{74.8/(13-3)} = 108.17 > 4.0.$$

Therefore x_1 enters the model.

Before going to the next step, we need to check if x_4 , which entered at Step 1, should be removed from the model. So we calculate its F_{out} statistic:

$$F_{\text{out}} = \frac{\text{SSE}(x_1) - \text{SSE}(x_1, x_4)}{\text{MSE}(x_1, x_4)} = \frac{1265.7 - 74.8}{74.8/(13 - 3)} = 159.20 > 4.0.$$

Therefore, x_4 cannot be removed from the model.

Step 3: Next, we check whether x_2 or x_3 should be added to the model. Toward this end, we calculate

$$\text{SSE}(x_1, x_2, x_4) = 47.97, \text{SSE}(x_1, x_3, x_4) = 50.84.$$

Since adding x_2 gives the smaller SSE, it is the candidate to enter the model next. To test if it meets the entry criterion, we calculate its F_{in} statistic:

$$F_{\text{in}} = \frac{\text{SSE}(x_1, x_4) - \text{SSE}(x_1, x_2, x_4)}{\text{MSE}(x_1, x_2, x_4)} = \frac{74.8 - 47.97}{47.97/(13 - 4)} = 5.034 > 4.0.$$

Therefore x_2 enters the model.

Before going to the next step, we need to check if either x_1 or x_4 should be removed from the model. So we calculate

$$\text{SSE}(x_1, x_2) = 57.9 \text{ and } \text{SSE}(x_2, x_4) = 868.9.$$

Thus, removing x_4 gives a smaller SSE. So we check if it can be removed by calculating its F_{out} statistic:

$$F_{\text{out}} = \frac{\text{SSE}(x_1, x_2) - \text{SSE}(x_1, x_2, x_4)}{\text{MSE}(x_1, x_2, x_4)} = \frac{57.9 - 47.97}{47.97/(13 - 4)} = 1.863 < 4.0.$$

Therefore we remove x_4 from the model. Recall that x_4 was the first variable to enter the model but it is removed at this step.

It is easy to check that neither x_1 nor x_2 can be removed from the model.

Step 4: Since x_4 was just removed from the model, it cannot re-enter (its F_{in} at this step equals its F_{out} at the previous step). Thus, it only remains to check whether x_3 can be entered in the model. We calculate $\text{SSE}(x_1, x_2, x_3) = 48.11$. Hence, its

$$F_{\text{in}} = \frac{\text{SSE}(x_1, x_2) - \text{SSE}(x_1, x_2, x_3)}{\text{MSE}(x_1, x_2, x_3)} = \frac{57.9 - 48.11}{48.11/(13 - 4)} = 1.832 < 4.0.$$

Therefore x_3 cannot enter the model and the algorithm stops with the final model consisting of two variables: x_1 and x_2 , which is the same model obtained using the best subsets regression with the C_p criterion.

It should be noted that stepwise regression in R uses the AIC criterion and not partial F-tests. Variables are added or removed from the model in a stepwise manner to minimize AIC. The following example illustrates this using the same Hald cement data.

Example:

(Hald Cement Data: Stepwise Regression Using R) Stepwise regression in R finds the model with the minimum AIC by deleting (in case of backward elimination) or adding (in case of forward selection) one variable at each step.

The resulting output of backward elimination algorithm

```
Start:  AIC=26.94 y ~ x1 + x2 + x3 + x4
```

	Df	Sum of Sq	RSS	AIC
- x3	1	0.1091	47.973	24.974
- x4	1	0.2470	48.111	25.011
- x2	1	2.9725	50.836	25.728
<none>			47.864	26.944
- x1	1	25.9509	73.815	30.576

```
Step:  AIC=24.97
```

```
y ~ x1 + x2 + x4
```

	Df	Sum of Sq	RSS	AIC
<none>			47.97	24.974
- x4	1	9.93	57.90	25.420
- x2	1	26.79	74.76	28.742
- x1	1	820.91	868.88	60.629

```
Call:
```

```
lm(formula = y ~ x1 + x2 + x4, data = cement)
```

```
Coefficients:
```

(Intercept)	x1	x2	x4
71.6483	1.4519	0.4161	-0.2365

- We see from the output that the algorithm starts with the full model, deletes one variable at a time and calculates the AIC of the resulting model.
- The results are arranged in the increasing order of AIC.
- The full model has $AIC = 26.94$.
- The best model of size three is $\{x_1, x_2, x_4\}$ and has $AIC = 24.97$.
- From the next step, we see that removing any more variables from this model increases AIC. So the overall best model is $\{x_1, x_2, x_4\}$.
- The same result is obtained if we run the forward stepwise algorithm.

Start: $AIC=71.44$

$y \sim 1$

	Df	Sum of Sq	RSS	AIC
+ x4	1	1831.90	883.87	58.852
+ x2	1	1809.43	906.34	59.178

+ x1	1	1450.08	1265.69	63.519
+ x3	1	776.36	1939.40	69.067
<none>			2715.76	71.444

Step: $AIC=58.85$

$y \sim x4$

	Df	Sum of Sq	RSS	AIC
+ x1	1	809.10	74.76	28.742
+ x3	1	708.13	175.74	39.853
<none>			883.87	58.852
+ x2	1	14.99	868.88	60.629
- x4	1	1831.90	2715.76	71.444

Step: $AIC=28.74$

$y \sim x4 + x1$

	Df	Sum of Sq	RSS	AIC
+ x2	1	26.79	47.97	24.974
+ x3	1	23.93	50.84	25.728
<none>			74.76	28.742
- x1	1	809.10	883.87	58.852
- x4	1	1190.92	1265.69	63.519

Step: AIC=24.97

$y \sim x4 + x1 + x2$

	Df	Sum of Sq	RSS	AIC
<none>			47.97	24.974
- x4	1	9.93	57.90	25.420
+ x3	1	0.11	47.86	26.944
- x2	1	26.79	74.76	28.742
- x1	1	820.91	868.88	60.629

Call:

`lm(formula = y ~ x4 + x1 + x2, data = cement)`

Coefficients:

(Intercept)	x4	x1	x2
71.6483	-0.2365	1.4519	0.4161

Choosing the Optimal Model

Best subset selection, forward selection, and backward selection each produce a collection of models, where each model contains a different subset of the p predictors.

- To use these methods, we need a way to decide which model is best.
- the model with all predictors will always have the smallest RSS and the largest R^2 , because these measures are based on training error.
- However, we want to choose a model with low test error.
- Training error is often a poor estimate of test error, so RSS and R^2 are not suitable for selecting among models with different numbers of predictors.

To select the model with respect to test error, we need to estimate test error. There are two common approaches:

1. Indirect estimation: Adjust the training error to account for the bias caused by overfitting.
2. Direct estimation: Use a validation set or cross-validation approach.

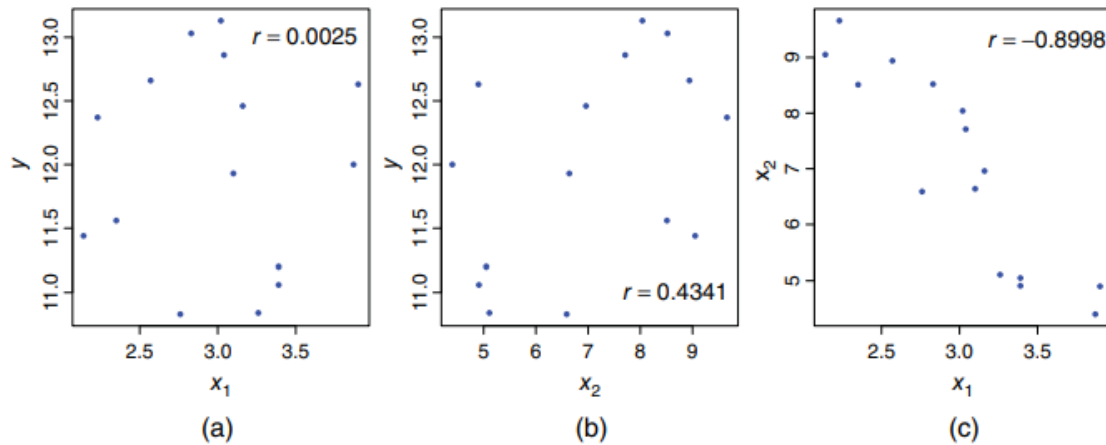
Exercises:

1. What is the Bias-Variance Trade-off?
2. How does bias affect model performance?
3. How does variance affect model performance?
4. Why does multicollinearity create problems in regression models?

5. Given a regression model that has high training accuracy but performs poorly on new data, is the model suffering from high bias or high variance?
6. If a model has high bias, what happens to its error on training and test data?
7. We applied forward stepwise regression to the Hald cement data, apply backward stepwise regression to the same data. You may use the SSE values from the example.
8. We perform best subset, forward stepwise, and backward stepwise selection on a single data set. For each approach, we obtain $p + 1$ models, containing $0, 1, 2, \dots, p$ predictors. Explain your answers:
 - a. Which of the three models with k predictors has the smallest training RSS?
 - b. Which of the three models with k predictors has the smallest test RSS?
 - c. True or False:
 - i. The predictors in the k -variable model identified by forward stepwise are a subset of the predictors in the $(k + 1)$ -variable model identified by forward stepwise selection.
 - ii. The predictors in the k -variable model identified by backward stepwise are a subset of the predictors in the $(k + 1)$ variable model identified by backward stepwise selection.
 - iii. The predictors in the k -variable model identified by backward stepwise are a subset of the predictors in the $(k + 1)$ variable model identified by forward stepwise selection.
 - iv. The predictors in the k -variable model identified by forward stepwise are a subset of the predictors in the $(k + 1)$ -variable model identified by backward stepwise selection.
 - v. The predictors in the k -variable model identified by best subset are a subset of the predictors in the $(k + 1)$ -variable model identified by best subset selection.
9. For the following data:

x_1	x_2	y	x_1	x_2	y
2.23	9.66	12.37	3.04	7.71	12.86
2.57	8.94	12.66	3.26	5.11	10.84
3.87	4.40	12.00	3.39	5.05	11.20
3.10	6.64	11.93	2.35	8.51	11.56
3.39	4.91	11.06	2.76	6.59	10.83
2.83	8.52	13.03	3.90	4.90	12.63
3.02	8.04	13.13	3.16	6.96	12.46
2.14	9.05	11.44			

- a. Calculate the correlation matrix between y , x_1 , and x_2 . Notice that the correlations between y and x_1 and y and x_2 are quite small. The following Figure shows three bivariate scatter plots between y , x_1 and x_2 .



What do you notice? Verify the joint relationship of y with x_1 and x_2 by running a regression.

- b. Why forward stepwise regression would fail but backward stepwise regression would work for this data set?

10. The following table lists the SSEs obtained by fitting different models involving three predictor variables, x_1, x_2, x_3 , based on a total of $n = 20$ observations. (The constant term is included in all models.)

SSEs for all possible models with three predictors

Variables in model	SSE_p	p	Error d.f.	MSE_p	$R^2_{adj,p}$	C_p	AIC_p
None	950						
x_1	720						
x_2	630						
x_3	540						
x_1, x_2	595						
x_1, x_3	425						
x_2, x_3	510						
x_1, x_2, x_3	400						

- Complete the table by filling in the values of p , error d.f., MSE_p , $R^2_{adj,p}$, C_p , and AIC_p statistics.
 - Which models will be selected as the best using the $R^2_{adj,p}$, C_p , and AIC_p criteria? Which model will you choose and why?
 - Suppose that stepwise regression is to be carried out with $f_{in} = f_{out} = 4.0$. Which variable would be the first to enter the model? What is its F_{in} value?
 - Which will be the second variable to enter the model? What is its F_{in} value? What is its partial correlation coefficient with respect to y controlling for the first variable that entered the model?
 - Will the first variable that entered the model be removed upon the entry of the second variable? Check by doing the partial F -test.
 - Will stepwise regression enter the third variable in the model, i.e. will it choose the full model? Check by doing the partial F -test.
11. From a data set on 33 adult males on their systolic blood pressure (y) and three predictor variables: age (x_1), smoking history (x_2), and body mass index (x_3), the following correlation matrix was computed:

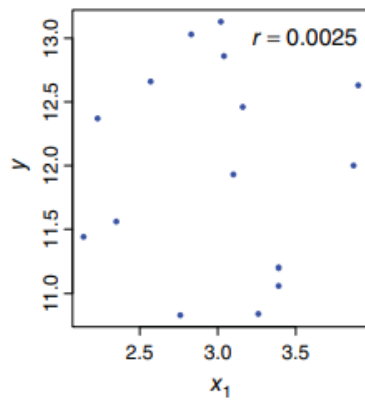
$$\begin{bmatrix} y & x_1 & x_2 & x_3 \\ 1 & 0.775 & 0.247 & 0.742 \\ 0.775 & 1 & -0.140 & 0.803 \\ 0.247 & -0.140 & 1 & -0.071 \\ 0.742 & 0.803 & -0.071 & 1 \end{bmatrix}.$$

- Stepwise regression chooses x_1 at Step 1 because it has the highest correlation with y . Which variable will be chosen next $-x_2$ or x_3 ? Determine by calculating $r_{yx_2|x_1}$ and $r_{yx_3|x_1}$.
- Explain why $r_{yx_2|x_1} > r_{yx_3|x_1}$ even though $r_{yx_2} < r_{yx_3}$?
- Do an F -test on $r_{yx_2|x_1}$ at $\alpha = 0.05$ to check if x_2 should be added to the model that already includes x_1 .

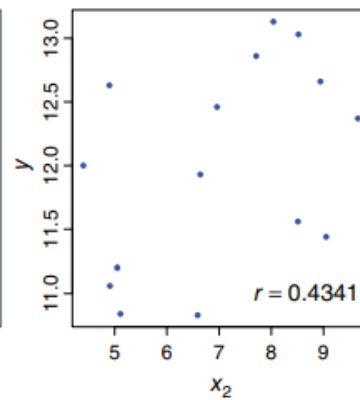
12. For the following data Calculate the correlation matrix between y, x_1 , and x_2 .

x_1	x_2	y	x_1	x_2	y
2.23	9.66	12.37	3.04	7.71	12.86
2.57	8.94	12.66	3.26	5.11	10.84
3.87	4.40	12.00	3.39	5.05	11.20
3.10	6.64	11.93	2.35	8.51	11.56
3.39	4.91	11.06	2.76	6.59	10.83
2.83	8.52	13.03	3.90	4.90	12.63
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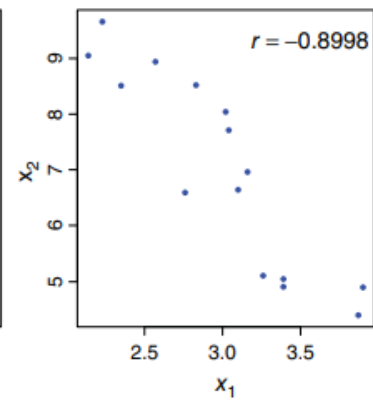
- The following Figure shows three bivariate scatter plots between y, x_1 and x_2 .



(a)



(b)



(c)

- What do you notice?
 - Verify the joint relationship of y with x_1 and x_2 by running a regression.
13. A real estate company wants to predict house prices using features like square footage, number of rooms, and location scores. They test two models:
- Model A: A simple linear regression with fewer features.
 - Model B: A polynomial regression (degree 6).
- a. Which model has high bias and low variance?
- b. Which model has low bias and high variance?
- c. How can the company improve model generalization?
14. A retail company builds a model to predict sales revenue using:
- X_1 : TV Advertising Spend
 - X_2 : Social Media Advertising Spend
 - X_3 : Online Advertising Spend
 - The overall F-statistic is significant, but individual predictor coefficients are insignificant.
- a. What statistical issue is present in the data?
- b. How can the company detect this issue?
- c. What are two solutions to fix the problem?